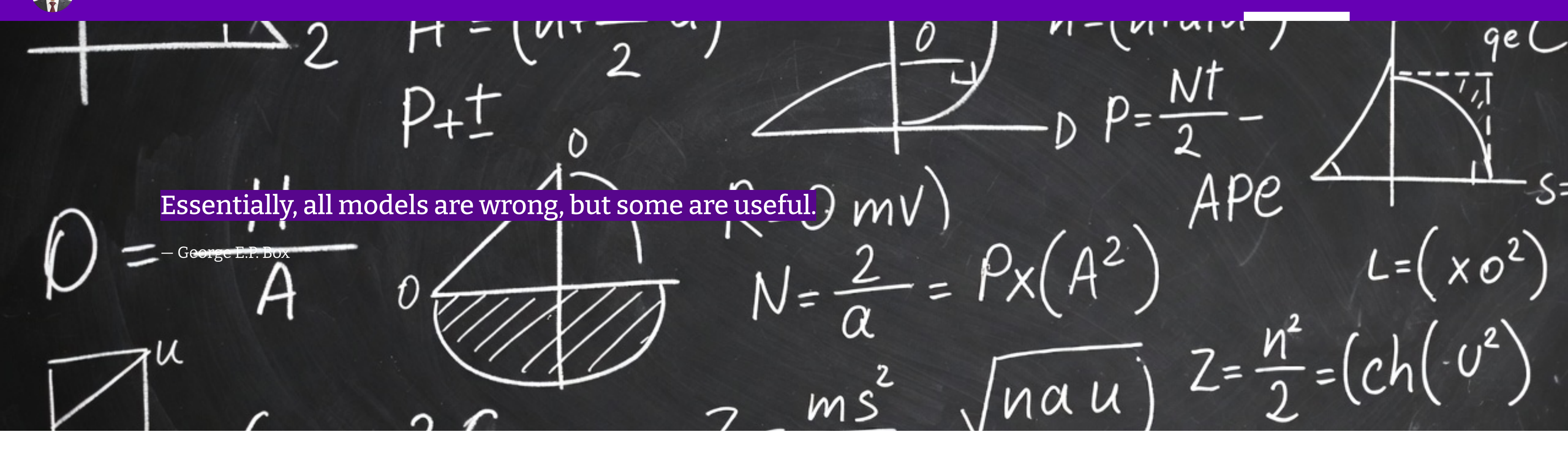

Essentially, all models are wrong, but some are useful.

— George E.P. Box

DATA SCIENCE & MACHINE LEARNING

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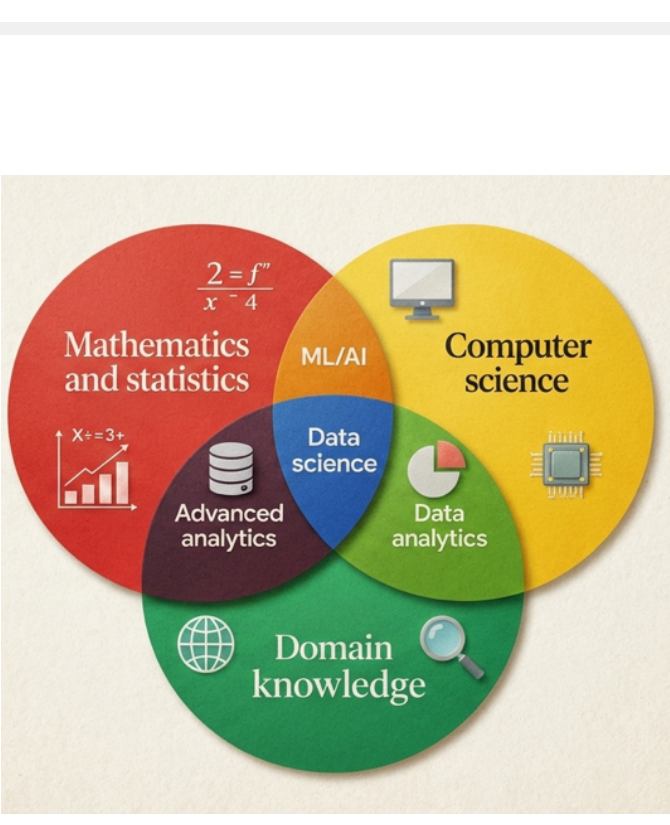


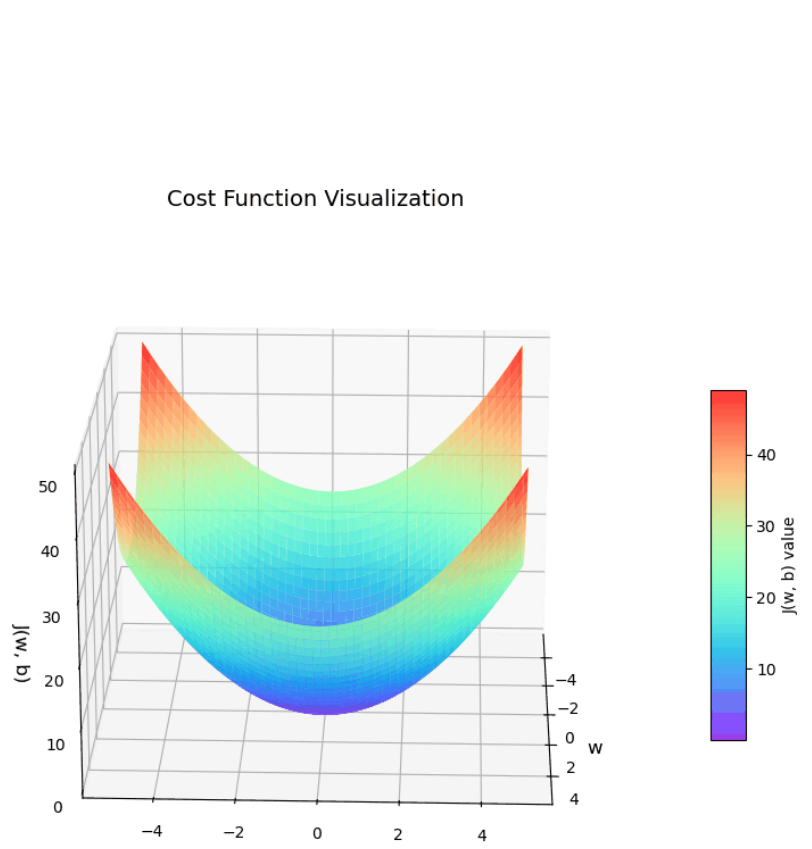
Developing practical data science and machine learning skills through applied projects

Data science is the multidisciplinary "detective work" of the digital age. It involves extracting meaningful insights from raw data to solve complex problems and predict future trends. It combines scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data.

Think of it as a bridge between several powerful domains:

- **Statistics & Mathematics:** The backbone of data science. It provides the tools to quantify uncertainty, identify patterns, and ensure that the "discoveries" made aren't just random flukes.
- **Computer Science:** This provides the horsepower. Data scientists use programming (like Python or R) and software engineering principles to process massive datasets that a human could never sort through manually.
- **Machine Learning (ML):** A subset of AI where computers "learn" from data. Instead of being explicitly programmed for every scenario, ML algorithms use statistical models to improve their performance on a specific task over time.
- **Deep Learning (DL):** A specialized branch of ML inspired by the human brain's neural networks. It's the tech behind advanced feats like facial recognition and natural language processing, requiring massive amounts of data and computational power.
- **Materials Science:** In this field, data science is revolutionary. Researchers use it to predict how new materials will behave—such as finding a more efficient battery compound or a stronger alloy—without having to run thousands of expensive, time-consuming physical experiments.

By combining these fields, data science transforms messy, unstructured information into a strategic roadmap for innovation.



👉🏼 View the full project

Supervised Machine Learning: Regression and Classification

Overview of Problems

The project addresses core supervised learning challenges: **predicting continuous values (regression) and binary outcomes (classification).** In the regression component, the problem involves modeling relationships in multi-variable datasets, such as **predicting housing prices based on features like size and location, while handling issues like non-linearity and scaling. The classification component tackles binary decision-making, such as determining if a tumor is malignant or benign based on medical features, complicated by overfitting and non-linear decision boundaries.** Together, these simulate real-world predictive modeling scenarios where data must be processed, models trained, and performance optimized to avoid errors like high variance or bias.

Objectives of the Projects

The primary objectives are to:

- Implement linear regression for continuous prediction with multiple features, incorporating techniques to enhance model accuracy and efficiency.
- Develop logistic regression for binary classification, focusing on cost minimization and regularization to improve generalization.
- Bridge theory and practice by coding algorithms from scratch, evaluating model performance, and applying optimizations like gradient descent and feature engineering.
- Achieve a comprehensive understanding of supervised learning pipelines, enabling learners to build deployable models for prediction and classification tasks.

Methodology

The methodology follows a step-by-step, code-based approach in Python, executed in Jupyter Notebooks:

- **Data Preparation**: Load and preprocess sample datasets (e.g., housing or medical data), including feature scaling (normalization) and engineering (e.g., adding polynomial terms for non-linearity).
- **Model Implementation**: Code the hypothesis functions—linear for regression and sigmoid-activated logistic for classification. Compute cost functions (mean squared error for regression, log loss for classification).
- **Optimization**: Apply gradient descent to minimize costs, with vectorization for efficiency and learning rate tuning. For classification, incorporate L2 regularization to penalize complex models and detect overfitting via train-test splits.
- **Evaluation and Iteration**: Visualize results (e.g., cost convergence plots, decision boundaries), assess metrics (e.g., error rates, accuracy), and iterate by adjusting hyperparameters. The process is progressive: regression builds foundational optimization skills, which are extended to classification's non-linear challenges. Total effort spans the graded labs, with ungraded labs providing preparatory practice.

Key Outcomes and Learning

- **Outcomes**: Successfully trained models with optimized parameters; for regression, achieved low mean squared error on test sets (e.g., via polynomial features reducing underfitting); for classification, attained high accuracy (e.g., 85-95% on balanced datasets) with regularization mitigating overfitting (e.g., improving test performance from overfitted baselines).
- **Learning**: Gained insights into why models fail (e.g., high variance from unregularized complexity) and how to fix them (e.g., feature scaling speeding convergence by 10x). Understood the importance of vectorization for scalable code and the transition from regression's linear predictions to classification's probabilistic outputs. Overall, reinforced iterative ML workflows, from hypothesis to deployment-ready models.

Skills and Tools

- **Skills**: Linear and logistic regression modeling, gradient descent optimization, vectorization, feature scaling and engineering, cost function derivation, overfitting diagnosis, regularization techniques, model evaluation (e.g., accuracy, precision), and visualization of learning curves/decision boundaries.
- **Tools**: Python for core implementation; NumPy for array operations, gradients, and vectorized computations; scikit-learn for optional validation and metric calculations; Jupyter Notebooks for interactive development and plotting.

Impact

This combined project equips learners with practical ML foundations, directly applicable to industries like healthcare (e.g., disease classification), finance (e.g., price prediction), and e-commerce (e.g., customer behavior modeling). It reduces common pitfalls in production ML, such as inefficient training or biased models, leading to more reliable AI systems. On a personal level, it builds a portfolio for career advancement in data science roles, fostering problem-solving confidence and ethical practices like emphasizing generalization for fair outcomes. Broadly, it contributes to the democratization of ML by enabling beginners to create impactful, real-world applications.

STAGE 2
22257

👉🏼 View the full project

Falcon 9 Booster Landing Success Prediction

Overview of the Problem

The core problem is a **binary classification task**: Predict whether the Falcon 9 first stage will land successfully (1) or not (0) based on historical launch data. SpaceX advertises Falcon 9 launches at ~$62 million, far below competitors' ~$165 million, largely due to reusable boosters. Accurate prediction of landing success helps estimate true launch costs and informs bidding strategies against SpaceX. The dataset covers SpaceX launches (primarily Falcon 9), including features like payload mass, orbit type, launch site, booster version, flight number, and more. Success rates have improved dramatically over time, making it an ideal case for analyzing trends and building predictive models.

Objectives of the Project

- Collect and prepare real-world SpaceX launch data using APIs and web scraping.
- Perform exploratory data analysis (EDA) to uncover patterns influencing landing success (e.g., payload mass, launch site, orbit).
- Create interactive visualizations and dashboards for insights and geospatial analysis.
- Build and evaluate multiple machine learning classification models to predict landing outcomes.
- Demonstrate full data science workflow proficiency: from data acquisition to model deployment insights, culminating in a portfolio-worthy project.
- Provide actionable business insights, such as factors enabling booster reuse and cost advantages.

Methodology

The project follows a structured, multi-stage data science pipeline across several Jupyter Notebooks:

- **Data Collection**: Use SpaceX REST API to fetch launch details; supplement with web scraping (e.g., Wikipedia or SpaceX site) for additional data like booster versions and outcomes.
- **Data Wrangling**: Clean data, handle missing values, create a binary landing outcome column (success/failure), apply one-hot encoding for categorical features (e.g., launch site, orbit), and standardize numerical features.
- **Exploratory Data Analysis (EDA)**: Use Pandas/NumPy for statistics; SQL queries (via sqlite or similar) for aggregations (e.g., success rates by site, payload ranges); visualizations with Matplotlib/Seaborn to explore trends over time, by site, payload, etc.
- **Interactive Visualizations**: Build Folium maps for launch site locations and success proximity; create Plotly Dash interactive dashboards for dynamic filtering and exploration.
- **Predictive Modeling (Classification)**: Train and compare models including Logistic Regression, Support Vector Machine (SVM), Decision Tree, K-Nearest Neighbors (KNN); use cross-validation, hyperparameter tuning (e.g., GridSearchCV), and metrics like accuracy, precision, recall, F1-score, confusion matrices.
- **Evaluation and Reporting**: Test on hold-out data; visualize decision boundaries/learning curves; compile findings into a presentation or report.

Tools are applied progressively, building on certificate modules.

Key Outcomes and Learning

- **Outcomes**: Models typically achieve **80-95% accuracy** on test data (commonly ~83-90% across learners, with Decision Tree often performing best due to interpretable splits on features like payload and orbit). Key insights include: higher success with certain payload ranges (~2000-4000 kg), specific sites (e.g., KSC LC-39A highest success), equatorial/proven orbits (e.g., GEO/SSO near 100% in subsets), and temporal improvements (success rate rising sharply post-2015/2016). Interactive dashboards and maps highlight geospatial and feature correlations effectively.
- **Learning**: Mastery of end-to-end DS pipeline; understanding feature importance (e.g., payload mass and booster type as strong predictors); appreciating model trade-offs (e.g., Decision Trees for interpretability vs. others for potential edge performance); importance of data quality, visualization for stakeholder communication, and business context in ML (cost prediction via reuse probability).

Skills and Tools

- **Skills**: Data collection (API + scraping), data wrangling/cleaning, EDA (statistical + visual), SQL querying, geospatial analysis, interactive dashboarding, binary classification, model selection/evaluation/tuning, hyperparameter optimization, interpretation of results, report/presentation.
- **Tools**: Python (core), Pandas & NumPy (data manipulation), Matplotlib/Seaborn (static plots), Folium (interactive maps), Plotly Dash (dashboards), Scikit-learn (ML models, preprocessing, metrics), Requests/BeautifulSoup (API/scraping), SQLite/Jupyter Notebooks (environment), optionally GitHub for portfolio.

Impact

This capstone creates a strong, tangible portfolio piece showcasing real-world DS application, highly valued for data science/analyst roles. It demonstrates ability to handle unstructured-to-structured workflows on aerospace data, relevant to space/tech industries. Insights into booster reuse factors highlight economic advantages of reusability, supporting competitive analysis in commercial spaceflight. Broader impact includes reinforcing ethical, data-driven decision-making in high-stakes domains like space exploration, while equipping learners to tackle similar predictive problems in other sectors (e.g., manufacturing reliability, predictive maintenance).

LEARNING PATH

Advancing my knowledge in Computational Materials Engineering for Smart Manufacturing Systems

I'm currently expanding my expertise in the following areas to integrate data-driven, simulation-based, and automation-focused approaches into my research:

👉 Data Science (DS) ➔ Machine Learning (ML) ➔ Deep Learning (DL)

👉 3D Modeling (CAD) ➔ FEA/Multiphysics Simulation (CAE) ➔ High Performance Computing (HPC)/Scientific Computing

👉 Laser Powder Bed Fusion (LPBF) or Selective Laser Melting (SLM), Direct Energy Deposition (DED), Binder Jetting, Multi-material AM, Cold spray AM, Fused Filament Fabrication or Fused Deposition Method (FFF/FDM), Stereolithography (SLA), Selective Laser Sintering (SLS), Robot-assisted AM

Explore my engineering capabilities

Shahadat Hussain, PhD

Researcher in Metamaterials, 3D Printing & Shape Memory Materials

Exploring advanced materials for innovative applications

Key Profiles

- LinkedIn
- Google Scholar
- ORCID
- ResearchGate
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- GitHub
- Grokipedia
- Institutional Page

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3D Printing

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- Raise3D Pro2 Plus
- Raise3D Forge1
- Ultimaker S1
- Stratasys F370
- Stratasys J850
- Formlabs Form3
- Bambu Lab Carbon X1

Design & Simulation

- FreeCAD
- Tinkercad
- MSLattice
- SolidWorks
- Materialise Magics
- IdeaMaker
- Cura
- Grabcad Print
- PreForm
- Bambu Studio

Data Science

- MATLAB
- Python
- Pandas
- Numpy
- Matplotlib
- Plotly
- Seaborn
- Scipy
- Scikit-learn
- SQLite
- Jupyter Notebook
- Linux
- Shell Scripting
- LaTeX

Mechanical Testing

- Instron 5969 (Bluehill Universal)
- Instron 8872 (WaveMatrix3)
- MTS 100 kN (TestSuite TW)
- Controls Simplex 350 & Automax (Multitest)
- KB 250 BVRZ (HardWin XL)

AVAILABILITY
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